To understand and compute the fidelity of a quantum state, fidelity serves as a measure of similarity between two quantum states, represented by density matrices ρ and σ. The general formula for fidelity is: Wikipedia +2

$$F(\rho, \sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right)^2$$

This expression, known as the Uhlmann fidelity, quantifies the closeness between two quantum states. For pure states, where at least one of the states is pure, the formula simplifies to: Wikipedia +1

$$F(\rho, \sigma) = \langle\psi|\sigma|\psi\rangle$$

where $|\psi\rangle$ is the pure state corresponding to ρ.

🔹 Example 1: Fidelity between two pure states

Let:

$$|\psi\rangle = \begin{bmatrix}1\\0\end{bmatrix}, \quad |\phi\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$$

Fidelity for pure states is:

$$F = |\langle\psi|\phi\rangle|^2$$

$$\langle\psi|\phi\rangle = \begin{bmatrix}1 & 0\end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}} \Rightarrow F = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

✅ **Fidelity = 0.5**

🔹 Example 2: Fidelity between a pure and a mixed state

Let:

- $|\psi\rangle = \begin{bmatrix}1\\0\end{bmatrix}$
- $\rho = \frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ (maximally mixed state)

Fidelity:

$$F = \langle\psi|\rho|\psi\rangle = \begin{bmatrix}1 & 0\end{bmatrix} \cdot \frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} \cdot \begin{bmatrix}1\\0\end{bmatrix} = \frac{1}{2}$$

✅ **Fidelity = 0.5**

🔹 Example 3: Fidelity between two mixed states

Let:

$$\rho = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.3 \end{bmatrix}, \quad \sigma = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.6 \end{bmatrix}$$

The fidelity is:

$$F(\rho, \sigma) = \left(\mathrm{Tr} \left[\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}} \right] \right)^2$$

Compute:

- $\sqrt{\rho} = \begin{bmatrix} \sqrt{0.7} & 0 \\ 0 & \sqrt{0.3} \end{bmatrix}$
- Multiply:

$$\sqrt{\rho} \cdot \sigma \cdot \sqrt{\rho} = \begin{bmatrix} \sqrt{0.7} & 0 \\ 0 & \sqrt{0.3} \end{bmatrix} \cdot \begin{bmatrix} 0.4 & 0 \\ 0 & 0.6 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{0.7} & 0 \\ 0 & \sqrt{0.3} \end{bmatrix} = \begin{bmatrix} 0.4 \cdot 0.7 & 0 \\ 0 & 0.3 \cdot 0.6 \end{bmatrix} = \begin{bmatrix} 0.28 & 0 \\ 0 & 0.18 \end{bmatrix}$$

- Now take the square root:

$$\sqrt{0.28} \approx 0.529, \quad \sqrt{0.18} \approx 0.424 \Rightarrow \mathrm{Tr} = 0.529 + 0.424 = 0.953$$

So,

$$F(\rho, \sigma) \approx (0.953)^2 \approx 0.908$$

✅ **Fidelity ≈ 0.908**